



Decoded: Fail Safe vs. Fail Secure – When and Where?

Understanding the basics:

- Fail safe products are unlocked when power is removed. Power is applied to lock the door.
- Fail secure products are locked when power is removed. Power is applied to unlock the door.
- Fail safe/fail secure refers to the status of the secure side (key side, outside) of the door.
- Most products provide free egress whether they are fail safe or fail secure (see below).

Electric strikes

An electric strike replaces the regular strike for a lockset or panic hardware. For a single door, it mounts in the frame and for a pair, it mounts in the inactive leaf or on a mullion. The lockset or panic

hardware still functions as it normally would—free egress is available at all times, except in the case of double-cylinder institutional function locks.

he spring-loaded
eeper on the
lectric strike
ontrols the
atchbolt of the
ock or panic
ardware. When
ccess is allowed
he keeper is free
nd the latchbolt
an be pulled
hrough the
eeper so the
oor can be
pened. When the
trike is secure,
he keeper
ecures the
atchbolt and
revents the door
om being
pened from the
utside. In most
ases, a key can
e used to retract
he latchbolt from
he secure side of
he door to allow
ccess if a
hanual override is
eeded. And
ecause the lock
r panic hardware
unctions
ndependently of
he electric strike,
ou can exit by
urning the lever
r pushing the
ouchpad of the
anic hardware,
egardless of
whether the
lectric strike is

ail safe or fail
ecure.

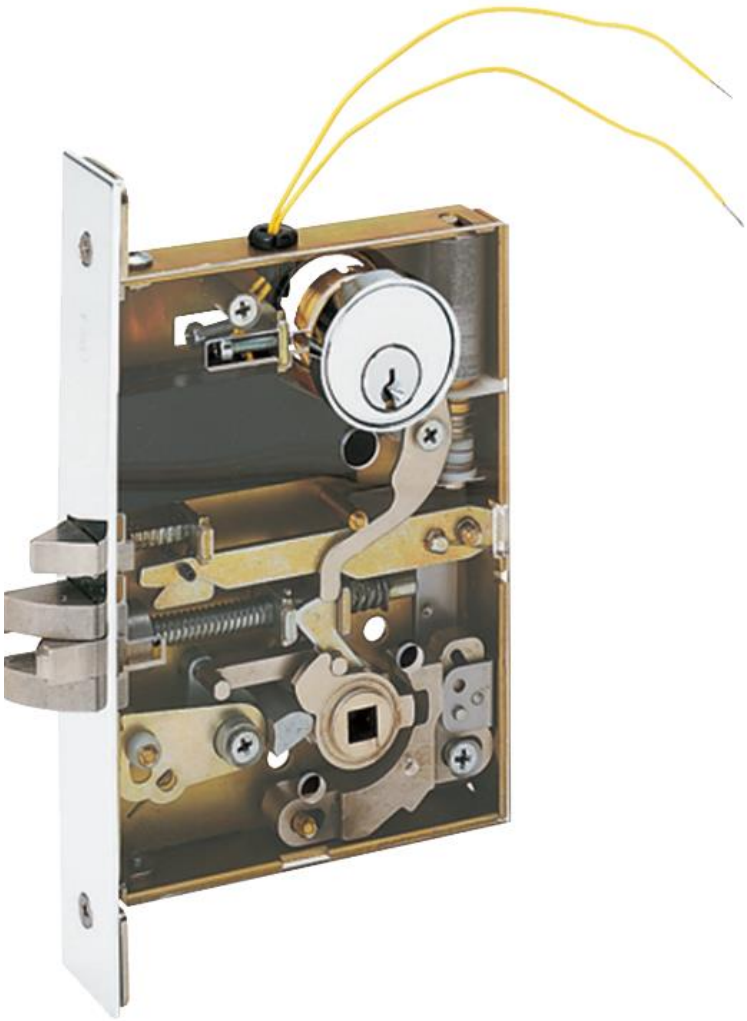
or electric strike
n fire-rated
oors, fail secure
trikes must be
sed because fail
afe strikes do nc
rovide the
ositive latching
nd the listings
equired by NFPA
0–Standard for
ire Doors and
ther Opening
rotectives. I
pecify fail secure
trikes in almost
ll applications,
xcept when
ccess is required
pon fire alarm.
here are very
mited situations
here access
pon fire alarm is
equired (see
elow regarding
tairwell re-entry)
have been asked
What about
refighter
ccess?" The use
f an electric
trike really
oesn't change
nything in regarc
o firefighter
ccess. Their
ethod for
ccess on a door
ith a mechanica
ockset can still b
sed. That might
e a key or
ccess-control
redential in the

ey box or a tool,
epending on
/hat type of
ardware is on th
oor.

You might think, “Let’s just make all electrified products fail safe so then I know there won’t be a problem.” Well, don’t forget that electric strikes on fire doors must be fail secure so the door is positively-latched if there is a fire. But in addition, there are security concerns. Should the building or area be unlocked and allow free access every time there is a power failure? A breach of security can be extremely dangerous for building occupants, along with the potential for loss or damage. That’s not a liability I’m willing to take on.

Electromechanical locks

An electromechanical lock is a lockset which has been electrified so that it can be controlled by a card reader, remote release or other access control device. Most electromechanical locksets allow free egress at all times. There are double-cylinder electromechanical locksets which do not allow free egress, just like a double-cylinder mechanical lockset, but neither of those should be used on any door that is required for egress. Note that when you see a lock Von Duprin® 6200 Series electric strike Schlage® L Series electrified mortise lock with two key cylinders, it may be a classroom security lock, which allows free egress, not an institutional function lock, which does not allow free egress.



Schlage® L Series electrified mortise lock

, fail secure electromechanical ockset is locked on the secure side hen there is no power to the lock. o unlock it, power is applied and he lever can then be turned to etract the latch. The latch remains rojected until the lever is turned.

, fail safe electromechanical ockset is locked when power is pplied and unlocked when power s removed. When power is emoved, the lever can be turned to etract the latch. Fail safe lectromechanical locks are used or stairwell doors providing re-ntry. The lock is constantly owered so that the lever on the tair side is locked. During a fire larm, the lever on the stair side is nlocked (power removed) either b he fire alarm or a signal from the re command center, depending or hich code has been adopted. uilding occupants may then leave he stair to find another exit if ecessary. The stair doors would lso be unlocked during a power ailure. The locks always allow free gress into the stair, with the xception of the stair discharge oor, which can be mechanically or lectrically locked on the outside ut allows egress out of the tairwell.

Electrified panic hardware trim

Electrified panic hardware trim refers to the outside lever on panic hardware or fire exit hardware. It operates the same way that an electromechanical lock does—the power controls whether the outside lever can be turned or not. The latch remains projected until the lever is turned, and free egress is always available by pushing the touchpad or crossbar of the panic hardware.

ail safe electrified
im for fire exit
ardware (panic
ardware listed for
se on fire doors) is
sed for stairwell
oors providing re-
ntry. Most other
oors are not
equired to allow
ccess upon fire
larm, so I typically
se fail secure
lectrified panic
ardware trim in
ocations other tha
tairwells. Keep in
nind that most
odes do not
equire the stair
ischarge door to
nlock upon
ctuation of the fire
larm. The door
etween the
tairwell and the
oof may be
equired, or desired
o be fail safe. This
s not typical and is
ot a requirement
f the International
uilding Code or
lFPA 101–The Life
afety Code. I have
nly worked on a
ew projects during
ny career where
ne path of egress
ed onto the roof.





Von Duprin® 99 Series panic hardware

Electric latch retraction

Electric latch retraction (EL/QEL) is a function typically used on panic hardware or fire exit hardware. EL devices, or QEL for the Von Duprin “Quiet” EL, are only available fail secure. When power is applied, the latch retracts automatically and stays retracted as long as power is applied. When power is removed, the latch is projected, securing the door. Again, free egress is provided via the touchpad of the panic hardware. EL/QEL devices are sometimes used on fire doors to allow push/pull function during normal use and provide positive latching during a fire alarm. A signal from the fire alarm system to the power supply is needed.

EL devices are often used with automatic operators, so the latch is retracted before the door begins to open. Electric strikes can perform this function as well. Fail safe or fail secure products can be used in this application, but I typically use fail secure except in the very rare case where access is required upon fire alarm. Electromechanical locks and electrified panic hardware trim are not used with automatic operators because the latch is not retracted until someone turns the lever, which would prevent the auto operator from opening the door.

Electromagnetic locks

An electromagnetic lock is an electromagnet which mounts on the frame with a steel armature mounted on the door. When you apply power to the magnet, it bonds to the armature, securing the door. Electromagnetic locks are only available fail safe. When you remove power, the electromagnetic lock unlocks.



Schlage® M400 Series electronmagnetic

ecause mag-
ocks do not
rovide free
gress like othe
lectrified
ardware,
elease devices
re required by
ode in order to
llow egress. A
lectromagneti
ock that is
eleased by
oor-mounted
ardware, like a
equest-to-exit
witch in panic
ardware, is
equired to
nlock upon
oss of power. I
ne
lectromagneti
ock is released
y a sensor, it
nust also
nlock upon
ctuation of a
ush button
ocated beside
ne door, upon
ctuation of the
re alarm or
prinkler systen
nd upon loss
f power.

Summary

- Fail safe locks should be used on stairwell doors requiring re-entry and any other doors which must allow free access upon fire alarm or power failure.
- Fail safe electric strikes can't be used for stairwell re-entry because fire doors require fail secure electric strikes for positive latching. Fire doors do not require fail secure electric locks—only fail secure electric strikes.
- Be aware that when a fail-safe product is used, the door will be unlocked whenever there is a fire alarm or power failure, which is an obvious security risk.

- Electric latch retraction panic hardware is only available fail secure—the latch projects when power is removed.
- Electromagnetic locks are only available fail safe—there is no magnetic bond when power is removed.
- Fail secure products are more common than fail safe due to security concerns. Power consumption may also be an issue. Fail secure products provide security when there is no power applied.
- Most electrified products, with the exception of electromagnetic locks, allow free egress at all times, regardless of whether they are fail safe or fail secure.

About the Author

Lori Greene, DAHC/CDC, CCPR, FDAI, FDHI is the Manager of Codes & Resources for Allegion. For more information about this topic and to download a free reference guide on codes, visit iDigHardware.com/guide